YAMLess Case Study — pos-elera-sales-compliance

YAMLess: Generating Azure Infrastructure from Code

Case Study — krogertechnology/pos-elera-sales-compliance

What is YAMLess?

YAMLess is an **Infrastructure from Code (IFC)** tool. Given a code repository, it produces a complete, policy-validated Terraform workspace — with no configuration files, no schema to fill in, and no IaC expertise required from the developer.

The pipeline has four stages:

```
Repo on disk
    │
    ▼
┌─────────────────────┐
│  1. Static Analyzer │  Reads Dockerfile, pom.xml, .env files, Helm values,
│                     │  Spring Boot YAML, source code (Java @Value, env refs).
│                     │  Produces a typed AppProfile with confidence scores.
└─────────────────────┘
          │  (fires only when any field is low/medium confidence)
          ▼
┌─────────────────────┐
│  2. LLM Gap-Filler  │  Claude reads relevant source files and fills in what
│                     │  static analysis couldn't determine — via structured
│                     │  tool use only, no free-form parsing.
└─────────────────────┘
          │
          ▼
┌─────────────────────┐
│  3. Golden Resolver │  Loads golden-profiles/<resource>.json — per-env
│                     │  sizing standards, cost estimate, governance
```

```
gates.
                    └──────────────────────┘
                               │
                               ▼
                    ┌──────────────────────┐
                    │  4. Renderer         │  Jinja2 root templates + static Terraform
modules →
                    │                      │  ready-to-apply workspace. Outputs AKS + K8s
manifests.
                    └──────────────────────┘
                               │
                               ▼
                    ┌──────────────────────┐
                    │  5. Validator        │  checkov policy-as-code. Security posture is
enforced
                    │                      │  by static module files — the LLM cannot
affect it.
                    └──────────────────────┘
```

Applying YAMLess to pos-elera-sales-compliance

pos-elera-sales-compliance is a Kroger Technology Java Spring Boot microservice responsible for POS (point-of-sale) sales compliance. It integrates with the ELERA POS platform, Ping Identity for OAuth2/OIDC SSO, Kount for fraud detection, MongoDB for persistence, and RabbitMQ for messaging.

Command run

```
yamless generate https://github.com/krogertechnology/pos-elera-sales-
        compliance \
  --no-llm \
  --target aks \
  --out-dir ~/yamless/outputs/pos-elera-sales-compliance-aks-infra \
  --validate
```

Profile detected

Field	Detected value	Confidence	Notes
Runtime	`java`	high	`pom.xml` present at repo root
Runtime version	`25`	high	Detected from Dockerfile Java base image
Port	`8080`	high	Standard Spring Boot port
Shape	`http-service`	high	Spring MVC /

Field	Detected value	Confidence	Notes
			REST controller detected
Database	`null`	—	MongoDB not in known pom.xml artifact set; LLM gap-filler required
Cache	`true`	high	Redis dependency detected in pom.xml
Queue	`false`	—	RabbitMQ present in secrets but not in known artifact set
Kafka	`false`	—	Not detected
Storage	`false`	—	Not detected
Secrets	43 detected	high	Helm values + Spring YAML + source code scanned
Ingress	`external: true`, port `8080`	high	
Scale trigger	`http`	high	HTTP service shape

Output generated

```
pos-elera-sales-compliance-aks-infra/
├── profile.json
├── terraform/
│   ├── main.tf
│   ├── variables.tf
│   ├── outputs.tf
│   ├── providers.tf
│   ├── backend.tf
│   ├── governance.tf
│   └── modules/
│       ├── aks_cluster/
│       ├── aks_workload_identity/
│       ├── keyvault/
│       ├── redis/          ← detected dependency
│       ├── vnet/
│       ├── postgres/       ← generated but unused (database: null)
```

```
│       ├── mysql/          ← generated but unused
│       ├── storage/        ← generated but unused
│       ├── servicebus/     ← generated but unused
│       └── event_hubs/     ← generated but unused
└── k8s/
    ├── namespace.yaml
    ├── service-account.yaml
    ├── secret-provider-class.yaml
    ├── deployment.yaml
    ├── service.yaml
    ├── ingress.yaml
    ├── hpa.yaml
    └── network-policy.yaml
```

35 .tf files + 8 K8s manifests written.

48/49 policy checks passed. One MEDIUM failure:

Check	Resource	Verdict
CKV_AZURE_43 — Storage account naming rules	`azurerm_storage_account.this`	False positive — `storage: false`; checkov scans all module files including unused ones

Cost estimate

Environment	Monthly	Annual
Dev	$160	$1,920
Staging	$310	$3,720
Production	$985	$11,820

Production breakdown:

Resource	Cost/month	SKU
AKS Cluster	$600	Standard, m5.large equivalent nodes
Redis Cache	$370	P1 (Premium — production standard)
Key Vault	$15	Standard
Total	**$985**	

Pricing as of 2026-11-06, approx. UK South on-demand.

Secrets detected (43)

Group	Count	Key examples
Ping Identity / OIDC	13	`PINGFED_CLIENT_SECRET`, `PING_AUTH_ISSUER_URI`, `PINGFED_DISCOVERY_URL`, `PING_CLIENT_ID`
MongoDB	2	`MONGODB_URI`, `SPRING_DATA_MONGODB_URI`

Group	Count	Key examples
RabbitMQ / AMQP	2	`SPRING_RABBITMQ_PASSWORD`, `AMQP_BROKER_PASSWORD`
Redis	2	`REDIS_HOST`, `REDIS_KEY`
ELERA POS	3	`ELERA_BASE_URL`, `ELERA_SERVICE_ACCOUNT_PASSWORD`, `ELERA_REDIRECT_URI`
Kount fraud	2	`KOUNT_API_KEY`, `KOUNT_BASE_URL`
Spring Cloud / Config	2	`CLOUD_CONFIG_URL`, `SWAGGER_CONFIG_URL`
Compliance APIs	4	`COMPLIANCE_BASE_URL`, `DB_URI`, `COM_KROGER_POS_COMPLIANCELOGGINGAPI_ENDPOINT`
Other	13	Auth timeouts, RSA keys, Grafana admin, loader catalog URLs

All 43 secrets are provisioned in Azure Key Vault and mounted to the pod via the Key Vault CSI driver.

What the output correctly captured

- AKS cluster with Workload Identity and Key Vault CSI wiring
- All 43 secrets surfaced into Key Vault — none left as plaintext env vars
- Redis Cache module (P1 in production, C1 in dev/staging)
- External HTTP ingress on port 8080 with Network Policy
- HPA (HTTP scale trigger, 1–10 replicas) + KEDA-ready ScaledObject
- Full K8s manifest set: Namespace, ServiceAccount, SecretProviderClass, Deployment, Service, Ingress, HPA, NetworkPolicy
- Governance gates in `governance.tf`: mandatory tags, min-replica policy, autoscaling, backup retention
- Security golden standards: RBAC-only auth, TLS 1.2, Key Vault firewall deny-all, private AKS nodes

What was missed

Missing	Root cause	Impact
MongoDB database module	`spring-boot-starter-data-mongodb` not in known pom.xml artifact set	No Cosmos DB API for MongoDB provisioned; app will fail to connect
RabbitMQ queue module	`spring-boot-starter-amqp` not in known artifact set	No Service Bus provisioned; AMQP messaging will fail
Port detection	No `EXPOSE` in Dockerfile (defaults to 8080 from shape detection)	Low risk — 8080 is correct for Spring Boot

Learnings for YAMLess

1. MongoDB support needed in Java artifact set

What happened: The pom.xml contains `spring-boot-starter-data-mongodb` and `de.flapdoodle.embed.mongo` (embedded test Mongo). The static analyzer's Java artifact table does not include MongoDB Spring Data as a database signal — it looks for Postgres/MySQL JDBC drivers and ORM packages.

Proposed fix: Add `spring-boot-starter-data-mongodb`, `mongodb-driver-sync`, and `flapdoodle.embed.mongo` to the Java dependency classifier. When detected, set `database: mongodb` and emit a Cosmos DB for MongoDB or Atlas module.

Impact: Affects any Java Spring Boot app using MongoDB — a common pattern in Kroger microservices.

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2. RabbitMQ / Spring AMQP detection missing

What happened: `spring-boot-starter-amqp` and `spring-rabbit` are present in `pom.xml`. The analyzer's queue detection covers Azure Service Bus SDK and `@azure/service-bus` for Node, but does not include Spring AMQP as a queue signal.

Proposed fix: Add `spring-boot-starter-amqp`, `spring-rabbit`, and `rabbitmq-client` to the queue classifier. When detected, set `queue: true` and emit a Service Bus module (the generated infrastructure equivalent of RabbitMQ in Azure).

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3. High secret count reflects complex external integration

The 43 detected secrets highlight the app's integration surface: PingFederate + PingOne (13 secrets), ELERA POS (3), Kount fraud (2), MongoDB (2), RabbitMQ (2). This is not a gap — the secret scanner correctly identified all of them from Helm values, Spring Boot YAML, and Docker Compose files. All are correctly provisioned in Key Vault.

However, the depth of Ping Identity integration (dual-IdP: PingFederate and PingOne simultaneously) is worth noting. The generated Kubernetes SecretProviderClass mounts all 43 secrets, which is unusually large and may warrant secret consolidation in a follow-up.

Improvement Roadmap

Priority	Change	Impact
High	Add MongoDB Spring Data to Java artifact classifier	Fixes `database: null` for this and similar Kroger Java microservices

Priority	Change	Impact
High	Add Spring AMQP / RabbitMQ to Java queue classifier	Generates Service Bus module for AMQP-based apps
Medium	Cosmos DB for MongoDB module (Azure)	Provides a native Azure-managed MongoDB-compatible store
Low	Secret consolidation recommendation when count > 30	Flag unusually large secret sets for review

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Summary

`pos-elera-sales-compliance` is a well-structured Java Spring Boot microservice with a clear HTTP service shape, correct port detection, and an exceptionally thorough secret inventory (43 secrets correctly identified). YAMLess produced a complete, policy-passing AKS workspace with 48/49 checkov checks passing.

The two gaps — MongoDB and RabbitMQ — are both static-analyzer limitations in the Java artifact classifier rather than fundamental tool constraints. Both would be resolved by the LLM gap-filler in a full run (`--no-llm` was used here). Once the artifact classifier is extended, YAMLess will generate the complete infrastructure for this service class automatically.

The $985/month production cost estimate is dominated by the AKS cluster ($600) and Premium Redis ($370), consistent with a POS compliance service requiring high availability and low-latency session caching.

Generated by YAMLess — https://github.com/zafar-ahmed-kr/YAMLess Date: 2026-11-10